Lecture-16 15/9/26

HVDC (High Voltage DC)

#Advantages

① It allows power transmission between two grids with different freq.

② Corona loss will be less in HVDC transmission than in HVAC transmission

③ HVDC requires less number of conductors than HVAC, which decreases the cost of transmission.

Types of HVDC Links:-

(i) Monopolar link: require only one conductor

(ii) Bipolar link: require two conductor

*HVAC require minimum three conductors.

④ HVDC transmission needs smaller towers than HVDC

⑤ AC is not uniformly distributed over the whole cross-section of the conductor, which is called the skin effect. Current tends to concentrate near the surface of the conductor. Electric current mainly flows at the skin of the conductor. DC has no skin depth also frequency is zero. So, [redacted] current is uniformly distributed over the conductor.

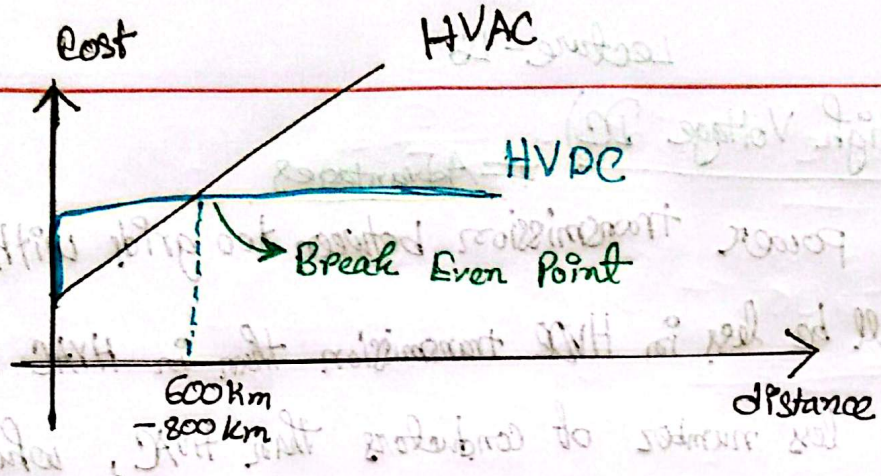
⑥ HVDC requires thinner conductors than HVAC, which reduces cost.

#Disadvantages.

① Terminal equipment cost of HVDC is very high than HVAC. The semiconductor converters are very expensive.

Generation → HVAC transmission → AC to DC converter → HVDC transmission → DC to AC converter → HVAC transmission → step down Tx for Distribution → Distribution

Electricity



* In Bangladesh HVDC is used when electricity is transmitted from India to Bangladesh.

TT-03 Probable date :- 27-09-26

Topic: HVDC, Stability, Mechanical Design of overhead line, Capacitance of Transmission Line

Presentation :- 21-9-26

[Power System II]

Lecture 22/09/26

Compensation in Power Systems

Art-15.2 Loading capability of Transmission Line :-

3 kinds of limitations

→ Dielectric limitations

→ Thermal capability

→ Stability issues.

Load Compensation : Mangement of ~~Reactance~~ Reactive power to improve power quality.

Reactive power flow controlled by shunt compensating devices (capacitors/inductors) at the end of the load (or at midpoint).

Art-15.4 Line compensation :- Modifying the characteristics of a line is known as Line compensation.

Various Compensating devices

→ Capacitors
→ Inductors } Connected as banks → Active/Passive compensator

→ Active voltage Sources (Synchronous generator) → Active compensator

① Shunt compensators
② Series compensators

Art-15.5 Series compensators

A capacitor in series with a line gives control over the effective reactance between line ends. The effective reactance is given by:

$$X_l' = X - X_C$$

X_C = Capacitor Reactance

This results in improvement in performance of the system as below:-

(i) Voltage drop in the line reduces (minimization of end-voltage variation)

(ii) Prevents voltage collapse.

(iii) Steady state power Transfer increases

(iv) Transient Stability increases.